

LEVEL 5 — Array Creation & Random Numbers

Easy line-by-line explanations with examples

1. `np.random`

`np.random` is NumPy's module for random-number generation.

```
import numpy as np
print(np.random.rand())
```

2. `np.random.rand()`

Generates random floats from 0 (inclusive) to 1 (exclusive).

```
arr = np.random.rand(3)
print(arr)
```

For 2 rows and 3 columns: `np.random.rand(2, 3)`

3. `np.random.randn()`

Generates values from the standard normal distribution (mean 0, standard deviation 1).

```
arr = np.random.randn(5)
print(arr)
```

4. `np.random.randint()`

Generates random integers; the upper limit is excluded.

```
arr = np.random.randint(1, 10, 5)
print(arr)
```

This gives 5 integers from 1 to 9.

5. `np.random.random()`

Generates random floats from 0 (inclusive) to 1 (exclusive).

```
arr = np.random.random(4)
print(arr)
```

6. `np.random.uniform()`

Generates random floating-point values in a chosen range.

```
arr = np.random.uniform(10, 20, 5)
print(arr)
```

7. `np.random.normal()`

Generates values from a normal distribution. The arguments are mean, standard deviation, and size.

```
arr = np.random.normal(50, 10, 5)
print(arr)
```

50 = mean, 10 = standard deviation, 5 = number of values.

8. `np.random.choice()`

Randomly selects values from a sequence.

```
arr = np.array([10, 20, 30, 40])
x = np.random.choice(arr)
print(x)
```

For 2 selections: `np.random.choice(arr, 2)`

9. `np.random.shuffle()`

Shuffles the original array in place.

```
arr = np.array([1, 2, 3, 4, 5])
np.random.shuffle(arr)
print(arr)
```

10. `np.random.permutation()`

Returns a shuffled permutation without changing the original array.

```
arr = np.array([1, 2, 3, 4, 5])
new_arr = np.random.permutation(arr)
print(new_arr)
print(arr)
```

11. `np.random.seed()`

Sets the starting state of NumPy's legacy global random generator, making repeated runs reproducible.

```
np.random.seed(10)
print(np.random.randint(1, 10, 5))
```

Using the same seed and same code gives the same sequence.

12. `np.random.default_rng()`

Creates the modern NumPy random-number Generator.

```
rng = np.random.default_rng()
print(rng.random(5))
```

13. Random Generator

The object returned by `default_rng()` is a Generator. Its methods create random values.

```
rng = np.random.default_rng()
print(rng.integers(1, 10, 5))
```

14. Reproducible Random Numbers

Use the same seed with `default_rng()` to reproduce a sequence.

```
rng1 = np.random.default_rng(42)
print(rng1.integers(1, 100, 5))
rng2 = np.random.default_rng(42)
print(rng2.integers(1, 100, 5))
```

Both generators produce the same sequence.

Quick Revision

`rand()` → floats 0 to 1

`randn()` → standard normal values

`randint()` → random integers

`random()` → floats 0 to 1

`uniform()` → floats in a range

`normal()` → normal distribution

`choice()` → random selection

`shuffle()` → changes original array

`permutation()` → returns shuffled permutation

`seed()` → reproducibility for legacy global RNG

`default_rng()` → modern Generator